

Harshit Kumar Singh

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LinkedIn | GitHub | Portfolio | LeetCode

Professional Summary

Software Engineer with hands-on experience building scalable full-stack web applications using React, FastAPI and PostgreSQL. Skilled in REST APIs, JWT authentication, relational database design, clean backend architecture and product-focused development. Strong foundation in Data Structures and Algorithms with a passion for solving real-world engineering problems.

Technical Skills

Languages: Java, Python, JavaScript, C++, SQL, HTML5, CSS3
Frontend: React.js, Redux Toolkit, Tailwind CSS, React Router, Axios, Vite
Backend: FastAPI, REST APIs, SQLAlchemy, JWT Authentication, Pydantic
Databases: PostgreSQL, MySQL
Core Concepts: OOP, DSA, Database Design, REST APIs, RBAC, Authentication
Tools: Git, GitHub, Postman, VS Code, Alembic

Experience

Center for Railway Information Systems (CRIS) May 2026 – Jul 2026
Machine Learning Intern, Kolkata

- Developed Python-based data processing and prediction pipelines for railway operational datasets.
- Automated preprocessing, feature engineering and evaluation workflows to improve reproducibility.
- Wrote modular and maintainable Python code while collaborating using Git.
- Evaluated multiple prediction models and documented experimental results.

Projects

Bake N Bite | Full Stack Food Ordering Platform GitHub Repository | Live Demo
React • FastAPI • PostgreSQL • JWT • Redux Toolkit • Tailwind CSS

- Developed a scalable full-stack food ordering platform supporting customer and administrator workflows.
- Implemented JWT authentication, role-based authorization and secure REST APIs.
- Designed modular backend architecture using Repository Pattern and Service Layer.
- Built responsive ordering, wishlist, review and inventory management modules.
- Optimized PostgreSQL queries for product, customer and order management.

Crowd Prediction and Staff Allocation System GitHub Repository | Live Demo
React • FastAPI • Python • PostgreSQL • XGBoost • LightGBM

- Developed a full-stack application for predicting metro crowd density and recommending staff allocation.
- Built REST APIs integrating prediction models with an interactive React dashboard.
- Visualized passenger traffic trends and staffing recommendations.
- Improved forecasting through feature engineering and model optimization.

Education

VIT Bhopal University Expected 2027
B.Tech, Computer Science and Engineering (AI & ML)
CGPA: 8.69/10

Kendriya Vidyalaya Cossipore 2022
Class XII (PCM): 88%

Certifications

Cloud Computing | Applied Machine Learning